

Challenge (Habanero)

- Q1.** Factor the trinomial $x^2 + 5x + 6$.
(A) $x(x + 3) + 2(x + 3)$
(B) $(x + 2)(x + 3)$
(C) $(x + 1)(x + 6)$
(D) $(x + 4)(x + 2)$
(E) $(x + 5)(x + 1)$
- Q2.** A hiker is traveling a distance of x miles. If the distance traveled is $x^2 + 4x + 4$ and it can be factored to find the time taken, what is the factored form?
(A) $(x + 1)^2$
(B) $(x + 2)^2$
(C) $(x + 3)^2$
(D) $(x + 4)^2$
(E) $(x + 5)^2$
- Q3.** Factor the expression $2x^2 + 5x - 3$.
(A) $(2x - 1)(x + 3)$
(B) $(2x + 1)(x - 3)$
(C) $(2x + 3)(x - 1)$
(D) $(2x - 3)(x + 1)$
(E) $(2x + 3)(x + 1)$
- Q4.** The population of a certain animal species can be modeled by $x^2 + 2x - 6$. Factor this expression.
(A) $(x + 3)(x - 2)$
(B) $(x + 1)(x - 6)$
(C) $(x + 2)(x - 3)$
(D) $(x - 1)(x + 6)$
(E) $(x - 3)(x + 2)$
- Q5.** The fuel consumption of a vehicle can be represented by $3x^2 - 2x - 1$. Factor this expression.
(A) $(3x + 1)(x - 1)$
(B) $(3x - 1)(x + 1)$
(C) $(3x - 2)(x + 1)$
(D) $(3x + 2)(x - 1)$
(E) $(3x - 1)(x - 1)$
- Q6.** The cost of maintaining hiking trails can be represented by $2x^2 + x - 1$. Factor this expression.
(A) $(2x - 1)(x + 1)$
(B) $(2x + 1)(x - 1)$
(C) $(2x - 1)(x - 1)$
(D) $(2x + 1)(x + 1)$
(E) $(2x - 1)(x + 2)$
- Q7.** A group of hikers is planning a trip. The total distance to be traveled can be represented by $x^2 + 3x - 4$. Factor this expression.
(A) $(x + 4)(x - 1)$
(B) $(x + 1)(x - 4)$
(C) $(x + 2)(x - 2)$
(D) $(x + 3)(x - 1)$
(E) $(x - 3)(x + 1)$
- Q8.** The altitude of a hiker can be represented by $x^2 - 2x - 3$. Factor this expression.
(A) $(x - 3)(x + 1)$
(B) $(x - 1)(x + 3)$
(C) $(x + 1)(x - 3)$
(D) $(x - 2)(x + 1)$
(E) $(x + 2)(x - 1)$
- Q10.** A park ranger is monitoring the population of a certain animal species. The population can be modeled by $2x^2 - 5x - 3$. Factor this expression and use it to determine the population when $x = 2$. Show all work and explain the steps used.

Open Ended Questions (FRQ)

- Q9.** Factor the expression $x^2 + 4x + 3$ and explain the method used to factor it. Provide at least 2 different methods.

Chef's Tips

- Remind students to show all work and explanations for FRQs
- Allow students to use calculators for checking, but not for calculations
- Provide extra paper for students if needed